

Multispectral Land-Cover Classification with EuroSAT: Does Spectral Range Beyond RGB Matter?

1. Problem Formulation

EuroSAT (Helber et al., 2019) is a land-cover benchmark of 27,000 Sentinel-2 patches, each 64×64 pixels with 13 spectral bands, labelled into ten classes: AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River and SeaLake. The problem is single-label patch classification.

Sentinel-2 records bands the eye cannot see, in the near-infrared (NIR), red edge and short-wave infrared (SWIR). Do they buy better accuracy than a model that sees only red, green and blue? And if so, can I get most of that benefit more cheaply by handing spectral indices to an RGB model, rather than training on the full band stack?

2. Data Preprocessing

Loading. The patches are GeoTIFFs, read with rasterio. One quirk matters: EuroSAT stores B8A (narrow NIR) last, at file index 12, not at its canonical position 8. I reorder all 13 channels into the SSL4EO-S12 canonical order (Wang et al., 2023) before selecting any bands, preventing silently mismatched channels.

Channel selection. Two views are taken from the same cube.

- *RGB*: canonical indices [3, 2, 1], i.e. B04 (red), B03 (green), B02 (blue).
- *MS*: canonical indices [1–8, 11, 12], giving B02–B08, B8A, B11 and B12 — ten bands. I drop B01, B09 and B10: they describe atmospheric absorption and aerosol scatter, not the surface, so they hold no land-cover signal here.

Preprocessing. Digital numbers are divided by 10,000 for approximate surface reflectance, then each patch is resized to 256×256 and centre-cropped to 224×224 , the input size the backbone expects.

Augmentation. During training, I apply random horizontal/vertical flips and 90° rotations; at patch scale, a field is a field whichever way it is turned, so every orientation is valid.

Split. I split the 27,000 patches once by stratified sampling: 21,600 (80%) train and 5,400 (20%) held-out test, saved to data/splits.npz. A validation set (15% of train, 3,240 patches) is carved at runtime and never stored, so the test set stays untouched across experiments.

3. Features / Model Architecture

3.1 CNN backbone

Both modalities use ResNet-18 from TorchGeo (Stewart et al., 2022) with Sentinel-2 MoCo weights: SENTINEL2_RGB_MOCO (3 channels) for RGB, SENTINEL2_ALL_MOCO (13 channels) for MS.

The 13-channel first convolution will not take a 10-channel input. I replace it with a Conv2d(10, 64, 7, 7) initialised by slicing the pretrained weights to the ten kept bands (`new_conv1.weight =`

`pretrained_conv1.weight[:, KEPT_MS_IDX, :, :]);` the atmospheric slices are dropped, and the land-band slices kept, so each band still meets its own filter.

The classification head is a small two-layer MLP, $\text{Linear}(512 \rightarrow 256) \rightarrow \text{ReLU} \rightarrow \text{Linear}(256 \rightarrow 10)$. I fine-tune with separate learning rates for the backbone (1×10^{-4}) and head (1×10^{-3}), batch size 64, mixed precision, and early stopping on validation accuracy (patience 10). Training ran on a single RTX 4060 Laptop GPU (8 GB) under PyTorch 2.3.1.

3.2 Spectral index features (32-d)

Eight physics-based indices are computed from the raw patch reflectance:

Index	Formula	Sensitivity
NDVI	$(B08 - B04) / (B08 + B04)$	Vegetation density
NDRE	$(B08 - B05) / (B08 + B05)$	Crop type / red edge
NDWI	$(B03 - B08) / (B03 + B08)$	Open water
MNDWI	$(B03 - B11) / (B03 + B11)$	Water vs built-up
NDWI_Gao	$(B08 - B11) / (B08 + B11)$	Vegetation water content
NDBI	$(B11 - B08) / (B11 + B08)$	Impervious surfaces
BSI	$((B11 + B04) - (B08 + B02)) / \dots$	Bare soil
SAVI	$1.5 \times (B08 - B04) / (B08 + B04 + 0.5)$	Soil-adjusted vegetation

For each index, I take four statistics (mean, std, p10, p90), giving $8 \times 4 = 32$ features per sample. No atmospheric bands are used.

3.3 LightGBM fusion arms

I combine the 512-d global-average-pooled embedding from each fine-tuned backbone (head removed) with the spectral-index vectors into five feature sets for a LightGBM classifier, with early stopping on the carved validation set.

3.4 Model and Framework Choices

ResNet-18 is a controlled choice, not a maximal one. TorchGeo's SENTINEL2_RGB_MOCO and SENTINEL2_ALL_MOCO share architecture, MoCo pretraining and the SSL4EO-S12 corpus, and differ only in input bands, so the RGB-versus-MS comparison varies nothing but the spectral input. No larger backbone (ViT, Swin, a foundation model) offers an equivalent RGB/13-band twin, so one would reintroduce the confound the comparison removes; on a saturated benchmark, extra capacity also buys little. Fine-tuning a pretrained backbone rather than training from scratch suits the modest dataset; early stopping at epochs 9–13 confirms the features transferred well.

The two end-to-end CNNs answer the primary question: does spectral range beyond RGB help? The five LightGBM arms answer a second one the CNN comparison cannot: whether that benefit comes from the bands themselves (recoverable by computing indices for an RGB model) or from learning jointly across bands end-to-end (not recoverable afterwards). They are a controlled ablation, not an attempt to beat the CNNs: `indices_only` isolates the standalone spectral signal, `rgb_cnn_indices` tests whether indices offset an RGB model's band-blindness, and `ms_cnn_indices` whether they add

anything once a CNN already sees all bands. Holding the classifier and feature protocol constant makes any difference attributable to the feature set alone.

4. Experiments Performed

ID	Arm	Input features	Dim
E1	indices_only	Spectral indices only	32
E2	rgb_cnn	Fine-tuned RGB CNN	512
E3	rgb_cnn_indices	RGB CNN + spectral indices	544
E4	ms_cnn	Fine-tuned MS CNN	512
E5	ms_cnn_indices	MS CNN + spectral indices	544

5. Evaluation Results

Accuracy is reported with macro-F1 because EuroSAT is mildly imbalanced (≈ 400 – 600 test patches per class): macro-F1 weights every class equally and exposes weakness on smaller classes that accuracy can hide. Because the deep models sit near the ceiling, McNemar's paired test decides whether the small gaps between models are real or specific to this split.

CNN end-to-end (test set, $n = 5,400$)

Model	Test Acc	Macro-F1	Epochs	Best Val Acc
RGB CNN	97.80%	97.76%	23	98.55%
MS CNN	98.65%	98.62%	19	99.07%

LightGBM arms (test set, $n = 5,400$)

Arm	Dim	Test Acc	Macro-F1
indices_only	32	89.87%	89.44%
rgb_cnn	512	98.02%	97.96%
rgb_cnn_indices	544	98.19%	98.13%
ms_cnn	512	98.85%	98.83%
ms_cnn_indices	544	98.85%	98.83%

McNemar significance tests ($n = 5,400$)

These tests compare paired predictions of the LightGBM arms, not the end-to-end CNNs.

Comparison	χ^2	p	Significant
rgb_cnn vs ms_cnn	18.09	2.1×10^{-5}	Yes
rgb_cnn_indices vs ms_cnn	12.01	5.3×10^{-4}	Yes
rgb_cnn vs rgb_cnn_indices	3.37	0.066	No
rgb_cnn_indices vs ms_cnn_indices	11.78	6.0×10^{-4}	Yes

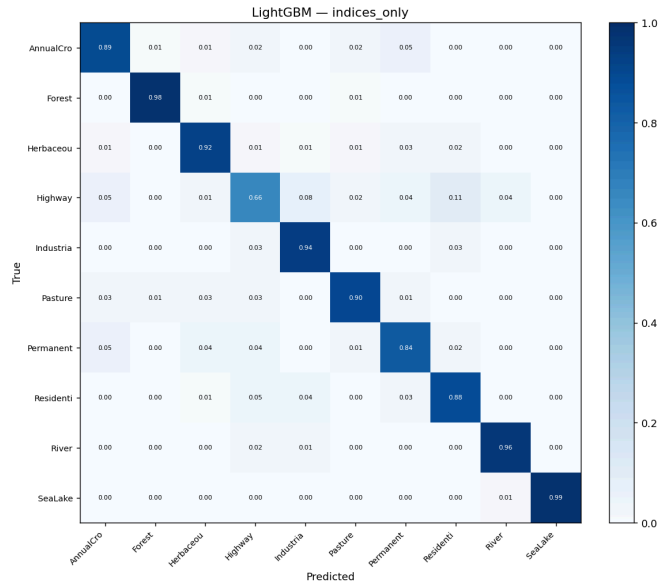


Figure 1 Indices-only confusion matrix.

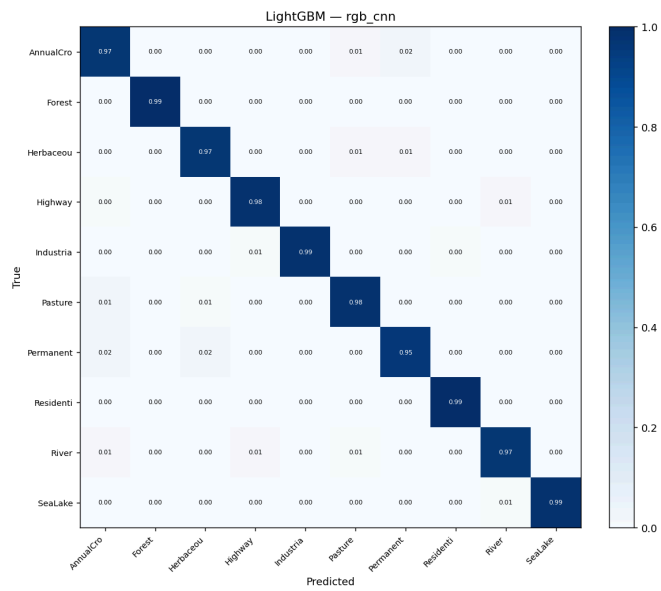


Figure 2 RGB CNN confusion matrix.

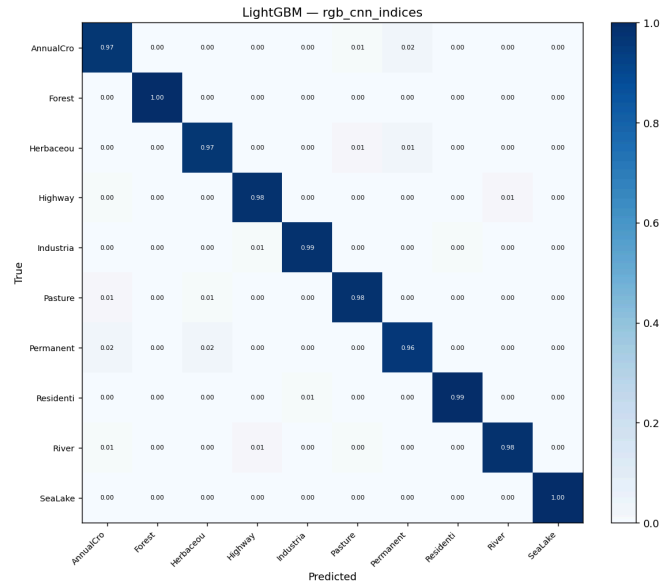


Figure 3 RGB CNN + indices confusion matrix.

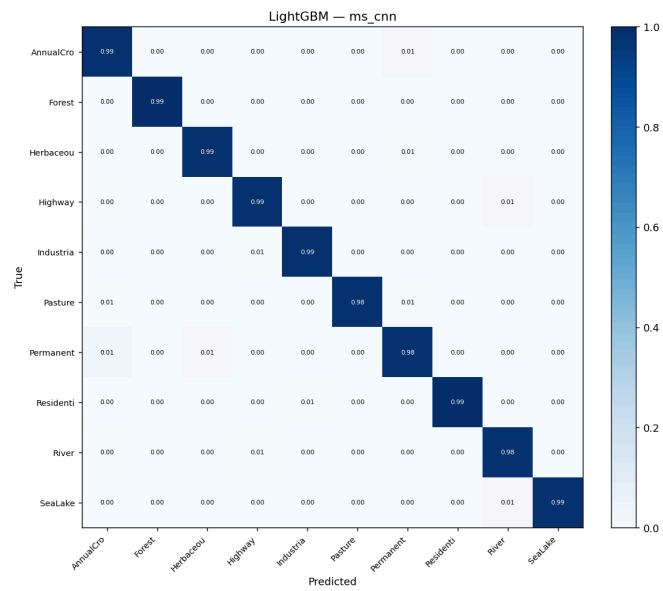


Figure 4 MS CNN confusion matrix.

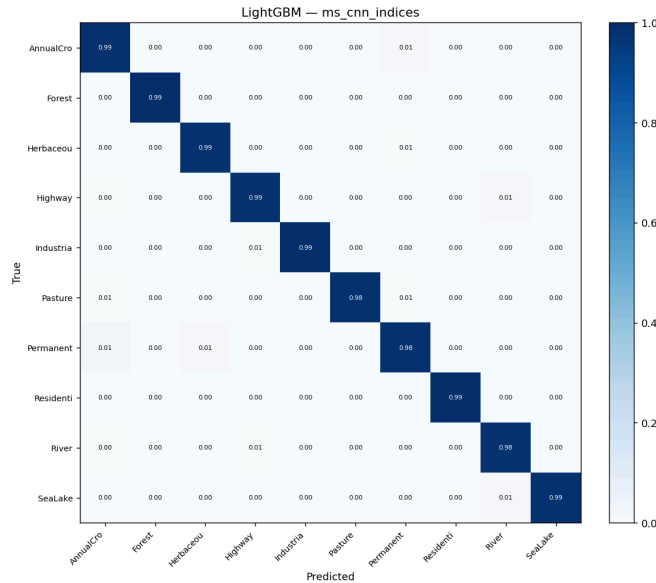


Figure 5 MS CNN + indices confusion matrix.

Per-class highlights (F1)

The MS gains over RGB land almost entirely on vegetation and crop classes: PermanentCrop 0.948→0.976, HerbaceousVegetation 0.963→0.981, AnnualCrop 0.968→0.980. SeaLake and Forest are already at the ceiling in both ($F1 \geq 0.986$). The indices-only model, by contrast, falls apart on Highway ($F1$ 0.706) and PermanentCrop ($F1$ 0.834) classes needing spatial layout or band contrasts; four statistics miss.

Confusion matrices, curves and galleries are under results/rgb/ and results/ms/; SHAP plots in results/ml/.

6. Key Results

1. **Multispectral input wins.** The MS CNN reaches 98.65% against the RGB CNN's 97.80%, a 0.85-point gain, and at the feature level the MS embedding beats the RGB embedding by a statistically significant margin (ms_cnn vs rgb_cnn, McNemar $p = 2.1 \times 10^{-5}$).
2. **Bolting indices onto RGB does not really close the gap.** The +0.17-point gain from rgb_cnn to rgb_cnn_indices is not significant ($p = 0.066$), and rgb_cnn_indices still trails ms_cnn by a significant margin ($p = 5.3 \times 10^{-4}$). Hand-crafted summary statistics are not a full replacement for training on the bands.
3. **For the MS model the indices are redundant.** ms_cnn_indices lands on the same 98.85% as ms_cnn; whatever the indices encode, within this feature protocol and test split, the indices provide no measurable incremental predictive value beyond the multispectral CNN embedding; this is consistent with redundancy.

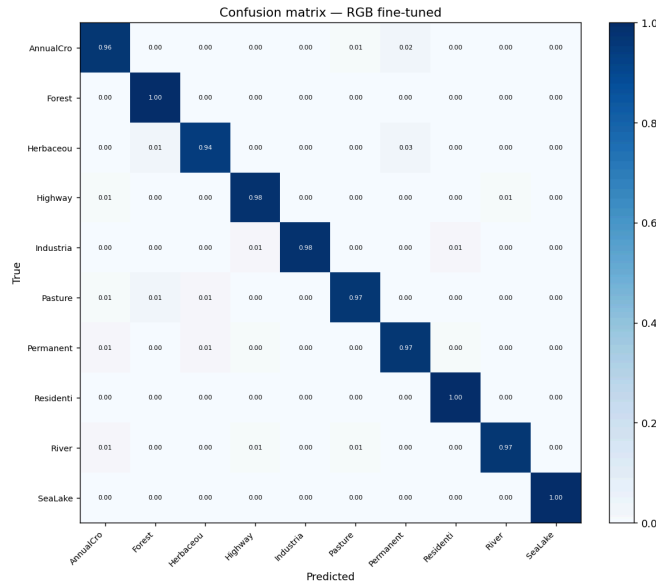


Figure 6 RGB CNN end-to-end confusion matrix.

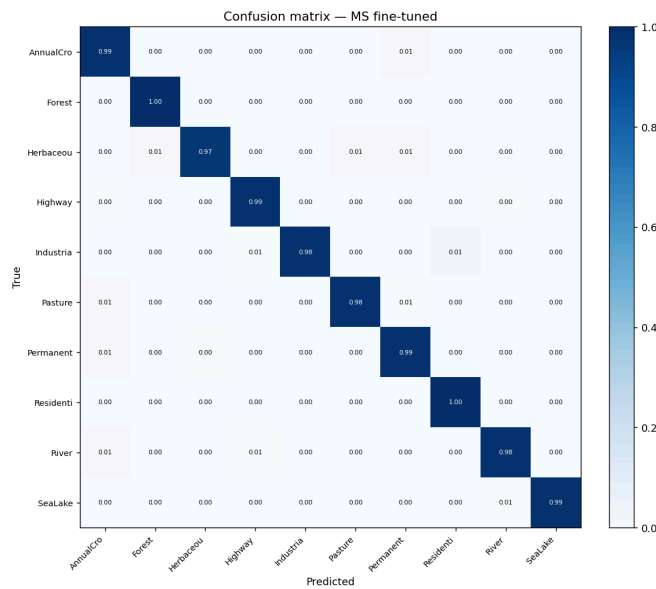


Figure 7 MS CNN end-to-end confusion matrix.

7. Discussion of Findings

The MS advantage is small in absolute terms, 0.85 points end-to-end, as expected on a benchmark this saturated. What matters is where it shows up. PermanentCrop, HerbaceousVegetation and AnnualCrop are the classic RGB headache: different crops and grasslands look almost identical in visible light but pull apart sharply in B05–B08 (red edge and NIR), which track canopy structure and chlorophyll. Forest and SeaLake, already solved in RGB, barely move: the extra bands matter where RGB is genuinely ambiguous and little elsewhere.

E3 asks whether indices can paper over the RGB model's blindness in the infrared. They mostly cannot. The +0.17-point gain points the right way: the indices do inject NIR/SWIR information the

CNN never saw, but it is not significant ($p = 0.066$), and a gap to the MS model remains. Two things explain that residual. First, reducing each index to four statistics throws away spatial arrangement: NDVI_p10 gives the least-vegetated pixel but not where it sits relative to a road or river, which the MS CNN learns across all ten bands at once and a 32-number vector cannot. Second, SHAP ranks MNDWI_std and NDWI_std highest, both measuring within-patch water/built-up heterogeneity, so the indices seem to help mainly at boundaries and mixed pixels, which a multi-band convolution already handles.

E5 settles the converse. Adding the same indices to the MS model changes nothing; ms_cnn_indices and ms_cnn both achieve 98.85% accuracy and 98.83% macro-F1. This indicates that, within this feature protocol and test split, the indices provide no measurable incremental value beyond the multispectral CNN embedding; this is consistent with the network learning similar spectral relationships. The contrast with E3 is the point: the indices carry only information that the RGB feature space is missing, and even there the gain is not significant ($p = 0.066$), whereas the MS space already contains it.

The gap between rgb_cnn_indices and ms_cnn_indices stays significant ($p = 6.0 \times 10^{-4}$): you cannot recover after the fact, with a few scalar summaries, what end-to-end training extracts from the bands. When NIR and SWIR are available, train on them rather than approximate them with indices.

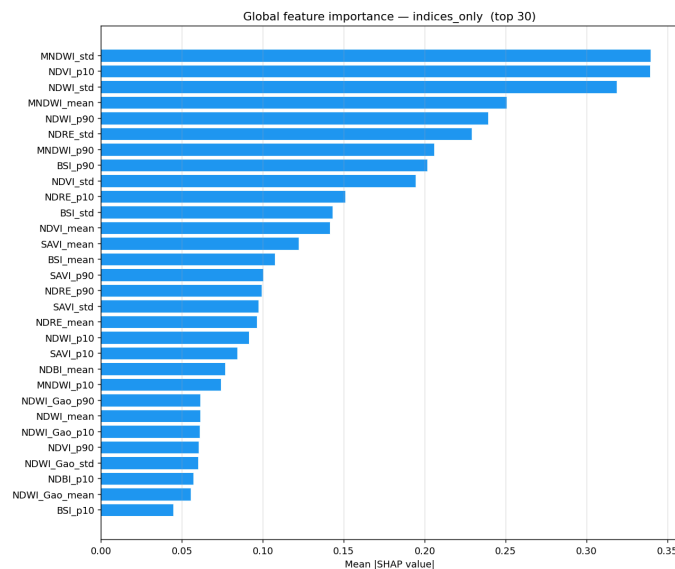


Figure 8. SHAP summary: MNDWI_std and NDWI_std dominate.

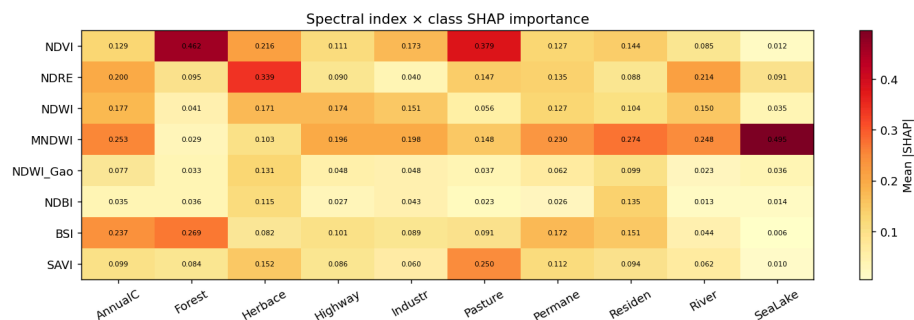


Figure 9. SHAP by class: NDVI and NDRE lead for crops.

Limitations and potential improvements: Results are based on one stratified random split of EuroSAT, so performance may be optimistic where geographically nearby patches share similar appearance; repeated splits and a geographically disjoint holdout would better measure transfer to new regions. EuroSAT also represents single-date observations, limiting discrimination between crop and vegetation types whose spectral signatures change across seasons. Resampling Sentinel-2 bands with different native spatial resolutions may reduce fine-scale red-edge and SWIR detail. Finally, the RGB and MS models use modality-specific pretrained weights, so the measured difference reflects both the additional bands and their multispectral pretraining.

References

1. Helber, P., Bischke, B., Dengel, A., & Borth, D. (2019). EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 12(7), 2217–2226.
2. Stewart, A. J., Robinson, C., Corley, I. A., Ortiz, A., Lavista Ferres, J. M., & Banerjee, A. (2022). TorchGeo: Deep learning with geospatial data. *Proceedings of the 30th International Conference on Advances in Geographic Information Systems (SIGSPATIAL)*.
3. Wang, Y., Braham, N. A. A., Xiong, Z., Liu, C., Albrecht, C. M., & Zhu, X. X. (2023). SSL4EO-S12: A large-scale multimodal, multitemporal dataset for self-supervised learning in Earth observation. *IEEE Geoscience and Remote Sensing Letters*, 20, 1–5.